AirClay: Few-Shot, Depth-Aware Gesture Control for 3D Sculpting in Blender

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1 Introduction

AirClay investigates whether an ordinary RGB webcam can support reliable, continuous two-hand control of a deformable mesh in Blender. The dominant hand pinches and drags to sculpt, the other hand controls the view, and a two-hand pinch changes the brush radius. The central difficulty is not recognizing a few isolated poses: natural motion, uncertain landmarks, occlusion, and transitions must not silently become unintended edits. A system with high frame accuracy can still be unusable if it frequently starts the wrong operation or releases a gesture prematurely.

Our proposal targets public-data pretraining, few-shot personalization, and calibrated camera-space displacement. This midterm establishes the prerequisite baseline: causal landmark processing, rule-based and learned intent interfaces, a shared interaction state machine, and Blender mesh control. The current sculpting motion is planar; neither user-adapted recognition nor camera-space depth has been validated. We separate implemented functionality, synthetic integration evidence, and planned human evaluation throughout this report.

In response to proposal feedback, the immediate research questions are whether a causal temporal model improves continuous intent recognition over static baselines, and whether filtering reduces unintended operations without excessive delay. Recorded-landmark replay provides a reproducible fallback for interactive testing. The existing evidence establishes a working software pipeline, not improved human recognition or successful few-shot, depth-aware sculpting.

2 Related Work

Continuous gesture recognition and landmarks. MediaPipe Hands combines palm detection and landmark estimation for RGB input (Zhang et al., 2020); it supplies our fixed front end, not task-specific intent labels. Jester provides short gesture clips (Materzynska et al., 2019), whereas IPN Hand includes uninterrupted gestures and intervening natural motion (Benitez-Garcia et al., 2021). IPN HandS adds frame-level skeleton annotations and refined gesture boundaries (Benitez-Garcia et al., 2025), making annotation quality relevant to temporal modeling. These datasets motivate reusable motion features, but their labels cannot simply be renamed as Blender commands. AirClay must also measure false operations, missed commands, and boundary delay rather than relying on isolated-clip accuracy.

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Clay motions and direct 3D interaction. SHREC 2024 studies seven classes of similar clay-molding motions using bilateral skeletal trajectories captured with Vicon (Veldhuijzen et al., 2024). It provides relevant two-hand motion structure, but its motion-capture inputs do not establish robustness to webcam noise or unintended edits in an interactive system. Mahdikhanlou and Ebrahimnezhad (2022) demonstrate webcam-based object deformation using colored gloves to track hand motion. AirClay instead uses bare hands and explicit operation boundaries in Blender. The research question is whether uncertain, continuous visual input can control edits reliably, rather than whether a virtual hand can be rendered.

Few-shot and user-defined gestures. Liu et al. (2022) combine MediaPipe landmarks, lightweight autoencoder features, and support-set similarity for few-shot gesture sequences. Schluesener and Buecker (2022) use LSTM-based Relation Networks with landmark sequences from a modified Jester dataset to recognize gesture sets from a few examples. These domain-specific approaches motivate adapting temporal representations to users instead of collecting a large command dataset for every person. Prototypical Networks (Snell et al., 2017) provide the general class-prototype formulation for our planned 1-, 5-, and 10-shot adaptation. We have not yet evaluated that transfer; support/query separation and rejection of natural non-command motion remain necessary.

Temporal architecture and signal stability. MLP and GRU remain static and recurrent baselines. Dilated causal TCNs offer a controlled alternative for sequence modeling (Bai et al., 2018); Transformers (Vaswani et al., 2017) are a later candidate requiring causal masking, suitable training data, and latency measurements. Neither architecture automatically provides usable pretraining. Interactive filtering research makes the jitter–lag tradeoff explicit (Casiez et al., 2012). In response to the proposal feedback, we will compare a causal TCN and None/EMA/Kalman filtering under fixed conditions, measuring operational errors alongside classification quality and delay. The 1 Euro filter motivates this tradeoff; it is not an implemented AirClay filter or a completed comparison.

Monocular depth and interacting hands. MobRecon targets efficient, temporally coherent single-view hand reconstruction (Chen et al., 2022); Neural Voting Field estimates camera-space hand geometry (Huang et al., 2023). ACR uses attention and cross-hand reasoning for arbitrary single- and two-hand configurations, including occlusion and truncation (Yu et al., 2023). These methods address geometry beyond hand-local articulation. Depth Anything V2 provides a general monocular depth alternative (Yang et al., 2024); its relative-depth model would still require calibration and hand-motion evaluation for our control task. RHD (Zimmermann and Brox, 2017) and InterHand2.6M (Moon et al., 2020) are planned sources of supervised single- and interacting-hand geometry. We retain these depth comparisons from the proposal as future work: current planar control and local MediaPipe coordinates do not establish shared camera-space depth or metric accuracy.

3 Methods

3.1 Implemented causal pipeline and baselines

The front end uses MediaPipe 0.10.21 with up to two hands and 21 landmarks per hand. Frames are mirrored once. Handedness and wrist continuity maintain left/right slots. The current tracker applies an exponential moving average (EMA), with $\hat{p}_t = 0.45p_t + 0.55\hat{p}_{t-1}$, only across gaps shorter than 150 ms. A causal 30 Hz resampler uses the most recent available observation and marks stale observations invalid; it does not interpolate using future frames.

For each hand, we concatenate 63 wrist-centered, palm-normalized local coordinates, two image-space wrist coordinates, two wrist velocities, a pinch ratio, and a presence indicator. Two-hand wrist distance and its velocity yield a 140-dimensional feature vector. Features are ordered by dominant/non-dominant role; left-hand chirality is canonicalized. Invalid hands contribute zeros with an absent presence indicator.

The rule baseline uses the image-space pinch ratio

$$r_t = \frac{\|u_{4,t} - u_{8,t}\|_2}{\max(\|u_{0,t} - u_{9,t}\|_2, 10^{-5})}, \qquad \mathcal{Y} = \{\text{idle}, \text{sculpt}, \text{orbit}, \text{resize}\}. \tag{1}$$

Pinch closes below 0.30 and opens above 0.45. The active hand role determines sculpt or orbit; two closed pinches request resize. The MLP uses the current frame's 135 non-velocity features, one

128-unit ReLU hidden layer, dropout 0.2, and four logits. The unidirectional GRU uses 24 frames, two 128-unit recurrent layers, dropout 0.2, and a linear head on the last output. Learned inference currently takes the largest logit and returns idle during window warm-up; probability smoothing and calibrated unknown-gesture rejection are not implemented.

Implemented training uses inverse-frequency weighted cross-entropy, AdamW at 10^{-3}, batch size 64, gradient clipping at 1, and validation Macro-F1 early stopping (patience 8, maximum 50 epochs). Existing trained weights are synthetic smoke-test artifacts, not human-trained models.

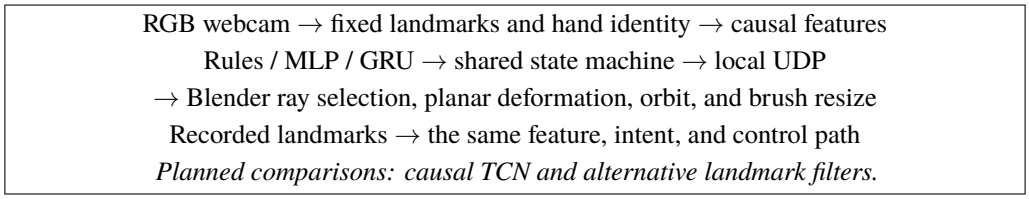


Figure 1: Implemented pipeline and replay fallback. Proposed comparisons are explicitly marked.

3.2 Interaction state machine and mesh control

All backends share a state machine with neutral re-arming, a 200 ms two-hand arbitration period, and a locked active mode. Other classes cannot steal an active operation. Rule-based release follows pinch hysteresis; learned release requires 100 ms of idle. Missing required hands pause the operation, and sustained loss for 300 ms cancels it. These mechanisms limit unintended changes but add response delay and do not correct a persistently wrong classification.

Local UDP packets carry session, sequence, operation, and control information. Blender selects the first visible surface with a view ray. A brush propagates over mesh edges using approximate geodesic distance, with displacement from an onset snapshot:

$$q_i = \min(d_i/R, 1), \qquad w_i = 1 - 3q_i^2 + 2q_i^3, \qquad v_i(t) = v_i(0) + w_i\Delta_t. \tag{2}$$

Here R is brush radius and Δ_t is view-plane displacement, capped at 0.5 Blender units. Orbit changes the view; resize changes the brush, not object scale. Only one interaction mode is active at a time. Completed operations enter a bounded undo/redo history; interrupted edits can be restored from their snapshots.

3.3 Human data and evaluation protocol (planned)

The pilot will include the five team members, each contributing two sessions. Each session targets 20 instances of each active intent and two minutes of natural motion. Prompt times supply rough labels only; a second member checks actual intent boundaries. With seed 42, participants will be split into three training, one validation, and one test subject before window generation. All sessions of a person stay in one partition. This is a small pilot with only one held-out test person, not a population-level generalization study.

3.4 Causal TCN comparison (planned)

The planned TCN will consume the same 24-frame, 140-dimensional sequence and predict the final frame's intent. Residual dilated convolutions will use left padding only; normalization must not aggregate future timesteps. The comparison will hold the front end, training split, state machine, and evaluation protocol fixed. No TCN implementation or measured advantage is claimed at this stage.

3.5 Filtering comparison (planned)

The existing EMA will be compared with no additional application-level smoothing and a constant-velocity Kalman filter. All variants must start from the same unsmoothed, identity-assigned observations. Each recording will reset filter state; missing detections must remain missing rather than become invented valid hands. The first comparison will hold the rule backend fixed to isolate the filter's effect. Image-space jitter will be reported in normalized image coordinates, not millimeters without calibration.

4 Results

4.1 Available implementation evidence

The Python regression suite passed all 57 tests in the October 19 audit. Historical synthetic Blender integration records demonstrate four committed operations (two sculpt edits, orbit, and brush resize), undo/redo, and timeout rollback on a 10,242-vertex mesh. One saved run reports a 16.48 ms median from message transmission to application. This excludes camera capture, landmark inference, intent waiting, and final display presentation; it is not end-to-end latency. MLP and GRU training, checkpoint loading, and evaluation completed two-epoch synthetic smoke runs. Those runs establish pipeline connectivity, not a meaningful ranking of recognition models.

4.2 Recognition and continuous events (not yet evaluated)

No human recognition results are available in the current evidence. Tables 1 and 2 reserve the planned comparison. Frame metrics will report raw classifier outputs, with state-machine outputs reported separately. Event metrics use post-controller labels: each ground-truth event is greedily matched to the unused same-class prediction with greatest positive overlap. Unmatched predictions are false operations; unmatched ground truth events are misses. Early release means a matched prediction ends over 100 ms early. Onset delay is predicted minus annotated onset and excludes unmatched events, so it must always be read alongside misses. Negative delays denote early activation.

Table 1: Planned human classification comparison. −−− means not measured. Learned metrics will be mean ± standard deviation across three seeds on the same held-out person; this variation does not estimate cross-user uncertainty.

	Macro-F1	Per-class recall				Model time
Method		Idle	Sculpt	Orbit	Resize	(ms/frame)
Rules	−−−	−−−	−−−	−−−	−−−	−−−
MLP	−−−	−−−	−−−	−−−	−−−	−−−
GRU	−−−	−−−	−−−	−−−	−−−	−−−
TCN (planned)	−−−	−−−	−−−	−−−	−−−	−−−

Table 2: Planned continuous-event comparison after the shared controller. Report missed and early-release event counts, not only delay of successful matches.

Method	False ops/min	Misses	Early releases	Onset (ms)	Offset (ms)
Rules	−−−	−−−	−−−	−−−	−−−
MLP	−−−	−−−	−−−	−−−	−−−
GRU	−−−	−−−	−−−	−−−	−−−
TCN (planned)	−−−	−−−	−−−	−−−	−−−

The recognition hypothesis remains untested. A useful comparison must establish whether temporal context helps transitions and ambiguous pinches, whether gains persist after the shared controller, and whether fewer false operations are caused by missed gestures rather than better rejection.

4.3 System timing and task completion (not yet evaluated)

The planned task is to create two protrusions, rotate the view, change brush radius, make another edit, undo once, and save the mesh. The five team members will receive equal practice and complete a rule-based pilot, with time, success, and unintended operations recorded. This is a usability check with familiar users, not a controlled comparison of learned models. The proposal targets at least 20 Hz updates and median camera-to-display latency below 250 ms; neither target has been established.

Table 3: System evidence and pending human measurements. Historical synthetic transport timing is not comparable to camera-to-display latency. −−− means unmeasured.

Measurement	Value	Evidence
Python regression tests	57 passed	Oct. 19 audit
Synthetic committed operations	4	Saved GUI run
Synthetic send-to-apply median	16.48 ms	Saved GUI run
Human camera-to-display median / p95	−−− / −−−	Pending
Human effective update rate	−−−	Pending
Task success / attempts; median duration	−−− / −−−; −−−	Pending
Ten-minute stability / recovery	−−−	Pending

5 Ablation Study (planned)

The primary ablation will compare filters with the rule backend and controller held fixed. For each participant, separately recorded 10 s stationary-hand intervals and repeated fast motion/release intervals will quantify jitter and response. For a stationary interval, image-space jitter is $J = \sqrt{N^{-1} \sum_t \|u_t - \bar{u}\|_2^2}$ for the index fingertip, using the same valid timestamps across filter variants. This measures repeatability, not absolute position accuracy. Filter tuning uses validation participants only; the principal held-out result uses the test person's intervals.

Table 4: Planned filter ablation on identical raw observations with the rule backend. EMA is implemented; the Kalman comparison and all human values remain pending.

Filter	Jitter (norm. image)	False ops/min	Misses	Onset (ms)
None	−−−	−−−	−−−	−−−
EMA ($\alpha = 0.45$)	−−−	−−−	−−−	−−−
Kalman (planned)	−−−	−−−	−−−	−−−

An additional offline-only ablation will remove the 200 ms onset arbitration while retaining the same EMA input, rule backend, mode locking, and loss handling. It will measure whether reduced onset delay increases mode errors or false operations. No mesh safety mechanism will be disabled in a live session for this experiment. Neither ablation has been measured; improved stability cannot yet be attributed to filtering or to the controller.

6 Discussion and Conclusion

The current evidence supports engineering feasibility: the feature, model, replay, communication, and mesh-editing components can be exercised together, with explicit operation boundaries and rollback. It does not establish robust human recognition. The principal remaining risks are landmark noise and identity errors, a small target dataset, latency accumulated across perception and arbitration, and uncalibrated depth.

Our intermediate goals respond directly to these risks: (1) replay verified recordings through the existing baselines; (2) compare a causal TCN and filtering under fixed conditions; and (3) validate real-camera interaction and stability. Recorded-landmark replay remains the fallback when live capture is unreliable. The proposed TCN is a controlled architectural comparison, not evidence that pretraining is available or that a more complex model will necessarily outperform a GRU.

Public-data pretraining, prototype adaptation at 1/5/10 shots, and calibrated depth remain future work. Public gesture labels must be reconciled with editor semantics, and few-shot support sequences must be separated from query sequences before windowing. Depth experiments will require calibration and independent camera-space ground truth; planar control remains the fallback. A later Transformer comparison will require a justified data source, causal inference, and a measured compute budget. Until the human tables are populated, no recognition, stability, or depth improvement over the baselines can be concluded.

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